

Bioinformatics: Practical 2

Name: Sidharthan S C

Roll no: BE21B039

1)

DBJ

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List of Entries

1 - 60 entries / Number of founds: 43461087 ☒ FlatFile ☐ XML ☐ Fasta View selected Download selected Download All

PrimaryAccessionNumber	Definition	SequenceLength	MolecularType	Organism
☐ L22430	Definition:Homo sapiens DNA sequence.	SequenceLength:470	MolecularType:DNA	Organism:Homo sapiens
☐ AC091799	Definition:Homo sapiens BAC clone RP11-416N13 from 7, complete sequence.	SequenceLength:48680	MolecularType:DNA	Organism:Homo sapiens
☐ AF147759	Definition:AF147759 Homo sapiens kidney fetal Homo sapiens cDNA clone EST96031830, mRNA sequence.	SequenceLength:488	MolecularType:mRNA	Organism:Homo sapiens
☐ AF147765	Definition:AF147765 Homo sapiens kidney fetal Homo sapiens cDNA clone EST96031802, mRNA sequence.	SequenceLength:517	MolecularType:mRNA	Organism:Homo sapiens
☐ AF147770	Definition:AF147770 Homo sapiens kidney fetal Homo sapiens cDNA clone EST, mRNA sequence.	SequenceLength:117	MolecularType:mRNA	Organism:Homo sapiens
☐ AF147775	Definition:AF147775 Homo sapiens kidney fetal Homo sapiens cDNA clone EST96031810-3', mRNA sequence.	SequenceLength:256	MolecularType:mRNA	Organism:Homo sapiens
☐ AF239918	Definition:AF239918 Homo sapiens liver 16 week fetus Homo sapiens cDNA, mRNA sequence.	SequenceLength:539	MolecularType:mRNA	Organism:Homo sapiens
☐ AF116186	Definition:AF116186 Homo sapiens placental skin Homo sapiens cDNA, mRNA sequence.	SequenceLength:264	MolecularType:mRNA	Organism:Homo sapiens
☐ AF147761	Definition:AF147761 Homo sapiens kidney fetal Homo sapiens cDNA clone EST96111222-3', mRNA sequence.	SequenceLength:362	MolecularType:mRNA	Organism:Homo sapiens
☐ AF147767	Definition:AF147767 Homo sapiens kidney fetal Homo sapiens cDNA clone EST96111229-3', mRNA sequence.	SequenceLength:427	MolecularType:mRNA	Organism:Homo sapiens
☐ AF147772	Definition:AF147772 Homo sapiens kidney fetal Homo sapiens cDNA clone EST96031860, mRNA sequence.	SequenceLength:196	MolecularType:mRNA	Organism:Homo sapiens
☐ AF147777	Definition:AF147777 Homo sapiens kidney fetal Homo sapiens cDNA clone EST96031825, mRNA sequence.	SequenceLength:781	MolecularType:mRNA	Organism:Homo sapiens
☐ AF005417	Definition:AF005417 Homo sapiens placenta Homo sapiens cDNA, mRNA sequence.	SequenceLength:265	MolecularType:mRNA	Organism:Homo sapiens
☐ AF498819	Definition:AF498819 Homo sapiens thyroid adenoma Homo sapiens cDNA clone 199, mRNA sequence.	SequenceLength:248	MolecularType:mRNA	Organism:Homo sapiens
☐ AF095909	Definition:AF095909 Homo sapiens thymus (Gastric N) Homo sapiens cDNA clone H425F1827676 similar to ACP7, melanin, mRNA sequence.	SequenceLength:361	MolecularType:mRNA	Organism:Homo sapiens

An official website of the United States government [Here's how you know.](#)

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National Library of Medicine

National Center for Biotechnology Information

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Nucleotide Search Help

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Species

Animals (28,130,855)

Plants (227,300)

Fungi (318,200)

Protists (175,583)

Bacteria (18,728,689)

Archaea (3,171)

Viruses (4,018,632)

Customize ...

Molecule types

genomic

DNA/RNA (38,818,283)

mRNA (10,233,993)

rRNA (1,306)

Customize ...

Source databases

INSDC (GenBank) (42,314,959)

RefSeq (9,384,351)

Customize ...

Sequence Type

Nucleotide (41,244,999)

EST (8,864,018)

GSS (1,795,947)

Genetic compartments

Chloroplast (5)

Mitochondrion (199,856)

Summary 20 per page Sort by Default order Send to Filters: [Manage Filters](#)

TAXONOMY

Was this helpful?

[Homo sapiens](#)

Human (*Homo sapiens*) is a species of primate in the family *Hominidae* (great apes).

Taxonomy ID: [9606](#)

[Genomes](#) [Genes](#) [Genome Data Viewer](#)

Results by taxon

Top Organisms [\[Tree\]](#)

Homo sapiens ([28434130](#))

Severe acute respiratory syndrome-related coronavirus ([8721809](#))

Escherichia coli ([4569025](#))

Klebsiella pneumoniae ([2334235](#))

Acinetobacter baumannii ([1595078](#))

All other taxa ([18592998](#))

More...

Find related data

Database:

Find items

Items: 1 to 20 of 64247275

<< First < Prev Page 1 of 3212364 Next > Last >>

☐ [Lynx pardinus genome assembly, contig: lp23s36493, whole genome shotgun sequence](#)

1. 4,643,072 bp linear DNA

Accession: CAAGRJ010006848.1 GI: 1603550370

[BioProject](#) [BioSample](#) [Protein](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens Kidd blood group protein \(SLC14A1\) gene, exons 4, 5 and partial cds](#)

Search details

"Homo sapiens"[Organism] OR Homo sapiens[All Fields]

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Text Search

Uses EBI Search to perform a free text search across ENA data. For more detailed usage please refer to the help & documentation section.

Search term:

Homo sapiens

Search

Search results for Homo sapiens

Assembly

Assembly (1,821)

Assembly View all 1,821 results.

GCA_002884475.1

YRI_HSA_v1 assembly for Homo sapiens

Sequence

Sequence (40,620,498)

Sequence (CON) (565,646)

Sequence (Standard) (40,054,852)

Sequence View all 40,620,498 results.

KJ946236

Homo sapiens Kidd blood group protein (SLC14A1) gene, exons 4, 5 and partial cds.

Sequence (CON)

View all 565,646 results.

KN707688

Homo sapiens unplaced genomic scaffold decoy00122.

Sequence (Standard)

View all 40,054,852 results.

KJ946236

Homo sapiens Kidd blood group protein (SLC14A1) gene, exons 4, 5 and partial cds.

Genome assembly contig set

View all 338,633 results.

CZVB02000000

Homo sapiens genome assembly

Transcriptome assembly contig set

View all 9 results.

GHIQ01000000

Homo sapiens, TSA project GHIQ01000000 data

Read

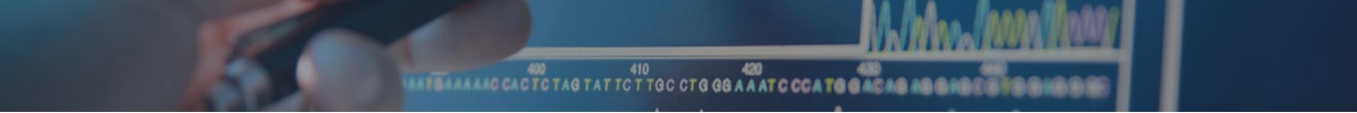
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DBJ has 43461087 entries which is less than GenBank: 64247275 while sequences deposited in EMBL is 40620 498.

2)

GC content is 45.681818%



Your input seq is:

ATGAGGAAGAAGCGGAGACGCGACGAAGAATGAAGAAGGTCCTTTGAAAGGTGTGAGTTGGCCAGAAGCTCTGAAAAGATTGGGAATGGATGGCTACAGGGGAATCAGCCTAGCAA>SEQ2ACTGGATGTGTTTGCCAATGGGAGAGTGGTTACAACACACGAGCTACAAACTACAATGCTGGAGACAGAGCACTGATTATGGGATATTTCA>SEQ3GATCAATAGCCGCTACTGGTGAATGATGGCAAAACCCAGGAGCAGTTAATGCCTGTCAATTTATCCTGCAAGTGCCTTTGCTGCAAGATAACATCGCTGATGCTGTAGCTGTGCAAGAGGGTTGTCCGTGATCCACAAGGCATTAGAGCATGGGTGGCATGGAGAAATCGTTGTCAAACAGAGGTGCCGTGATGTTCAGGTTGTGGAGTGAATGA

Nucleotide Content:

	Nucleotide content in %
AT_content	52.045455
Adenine_content	29.318182
Cytosine_content	16.363636
GC_content	45.681818
Guanine_content	29.318182
Keto_GT_content	52.045455
Purine_AG_content	58.636364
Thymine_content	22.727273

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3) GEN Bank

GenBank

Send to:

Change region shown

E. coli lac operator DNA fragment 5' to Z gene

GenBank: X03527.1

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DEFINITION

ACCESSION

VERSION

KEYWORDS

SOURCE

ORGANISM

REFERENCE

AUTHORS

TITLE

JOURNAL

PUBMED

COMMENT

FEATURES

ORIGIN

//

X03527

55 bp

DNA

linear

BCT 23-OCT-2008

E. coli lac operator DNA fragment 5' to Z gene.

X03527 M12479

X03527.1

operator; protein binding site.

Escherichia coli

[Escherichia coli](#)

Bacteria; Pseudomonadota; Gammaproteobacteria; Enterobacterales; Enterobacteriaceae; Escherichia.

1 (bases 1 to 55)

Manly,S.P. and Matthews,K.S.

lac operator DNA modification in the presence of proteolytic fragments of the repressor protein

J. Mol. Biol. 179 (3), 315-333 (1984)

6392562

On Mar 11, 2005 this sequence version replaced [M12479.1](#).

Location/Qualifiers

source

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[misc_feature](#)

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[repeat_region](#)

20..25

/note="inverted repeat A"

[variation](#)

24

/note="G is A in lac O-c"

[repeat_region](#)

25..30

/note="inverted repeat A"

[variation](#)

26

/note="G is T in lac O-c"

[variation](#)

27

/note="A is G in lac O-c"

[variation](#)

28

/note="G is T in lac O-c"

[variation](#)

29

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[variation](#)

30

/note="G is A in lac O-c"

[variation](#)

33

/note="T is C in lac O-c"

[variation](#)

36

/note="C is T in lac O-c"

1 cggctcgtat gttgtgtgga attgtgagcg gataacaatt tcacacagga aacag

//

Analyze this sequence

Run BLAST

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Related information

PubMed

Taxonomy

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E. coli lac operator DNA fragment 5' to Z gene

Nucleotide

Synthetic construct human lysozyme mRNA, complete cds

Nucleotide

Homo sapiens (64247275)

Nucleotide

GCF_000151735.1[assembly] NOT nuccore_comp_nuccore[filter] (3143)

Nucleotide

txid9606[Organism:exp] (1)

Genome

See more...

DDBJ

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DEFINITION

ACCESSION

VERSION

KEYWORDS

SOURCE

ORGANISM

REFERENCE

AUTHORS

TITLE

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FEATURES

ORIGIN

//

X03527

55 bp

DNA

linear

BCT 23-OCT-2008

E. coli lac operator DNA fragment 5' to Z gene.

[X03527](#) M12479

X03527.1

operator; protein binding site.

Escherichia coli

[Escherichia coli](#)

Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacterales; Enterobacteriaceae; Escherichia.

1 (bases 1 to 55)

Manly S.P., Shive M.K.

lac operator DNA modification in the presence of proteolytic fragments of the repressor protein

J. Mol. Biol. 179(3), 315-333(1984).

6392562

Location/Qualifiers

source

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27..27

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[variation](#)

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/note="G is T in lac O-c"

[variation](#)

29..29

/note="C is T in lac O-c"

[variation](#)

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/note="G is A in lac O-c"

[variation](#)

33..33

/note="T is C in lac O-c"

[variation](#)

36..36

/note="C is T in lac O-c"

16 a

9 c

16 g

14 t

1 cggctcgtat gttgtgtgga attgtgagcg gataacaatt tcacacagga aacag

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ID   X03527; SV 1; linear; genomic DNA; STD; PRO; 55 BP.
XX
AC   X03527; M12479;
XX
DT   02-JUL-1986 (Rel. 09, Created)
DT   23-OCT-2008 (Rel. 97, Last updated, Version 5)
XX
DE   E. coli lac operator DNA fragment 5' to Z gene
XX
KW   operator; protein binding site.
XX
OS   Escherichia coli
OC   Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacterales;
OC   Enterobacteriaceae; Escherichia.
XX
RN   [1]
RP   1-55
RX   DOI; 10.1016/0022-2836(84)90068-8.
RX   PUBMED; 6392562.
RA   Hanly S.P., Shive M.K.;
RT   "lac operator DNA modification in the presence of proteolytic fragments of
RT   the repressor protein";
RL   J. Mol. Biol. 179(3):315-333(1984).
XX
DR   MD5; 5dc87667dc65a72a21939dded7e3a887.
XX
FH   Key          Location/Qualifiers
FH
FT   source              1..55
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FT                       /db_xref="taxon:562"
FT   misc_feature        1..55
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FT   repeat_region       20..25
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FT   variation            26..26
FT                       /note="G is T in lac O-c"
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FT   variation            33..33
FT                       /note="T is C in lac O-c"
FT   variation            36..36
FT                       /note="C is T in lac O-c"
XX
SQ   Sequence 55 BP; 16 A; 9 C; 16 G; 14 T; 0 other;
    cggctcgtat gttgtgtgga attgtgagcg gataacaatt tcacacagga aacag
//

```

9)

EC Explorer

[browse EC tree] - [search]

EC 1.7.2.3 Details

show Brenda entry
EC number
1.7.2.3
Accepted name
trimethylamine- <i>N</i> -oxide reductase
Reaction
trimethylamine + 2 (ferricytochrome c)-subunit + H ₂ O = trimethylamine <i>N</i> -oxide + 2 (ferrocycytochrome c)-subunit + 2 H ⁺
Other name(s)
TMAO reductase, TOR, <i>torA</i> (gene name), <i>torZ</i> (gene name), <i>bisZ</i> (gene name), trimethylamine- <i>N</i> -oxide reductase (cytochrome c)
Systematic name
trimethylamine:cytochrome c oxidoreductase
CAS registry number
37256-34-1
Comment
Contains bis(molybdopterin guanine dinucleotide)molybdenum cofactor. The reductant is a membrane-bound multiheme cytochrome c. Also reduces dimethyl sulfoxide to dimethyl sulfide.
History
created 2002, modified 2018

EC Tree

- 1 Oxidoreductases
 - 1.1 Acting on the CH-OH group of donors
 - 1.2 Acting on the aldehyde or oxo group of donors
 - 1.3 Acting on the CH-CH group of donors
 - 1.4 Acting on the CH-NH₂ group of donors
 - 1.5 Acting on the CH-NH group of donors
 - 1.6 Acting on NADH or NADPH
 - 1.7 Acting on other nitrogenous compounds as donors
 - 1.7.1 With NAD⁺ or NADP⁺ as acceptor
 - 1.7.2 With a cytochrome as acceptor
 - 1.7.2.1 nitrite reductase (NO-forming)
 - 1.7.2.2 nitrite reductase (cytochrome; ammonia-forming)
 - 1.7.2.3 trimethylamine-*N*-oxide reductase
 - 1.7.2.4 nitrous-oxide reductase
 - 1.7.2.5 nitric oxide reductase (cytochrome c)
 - 1.7.2.6 hydroxylamine dehydrogenase
 - 1.7.2.7 hydrazine synthase
 - 1.7.2.8 hydrazine dehydrogenase
 - 1.7.2.9 hydroxylamine oxidase
 - 1.7.3 With oxygen as acceptor
 - 1.7.5 With a quinone or similar compound as acceptor
 - 1.7.6 With a nitrogenous group as acceptor
 - 1.7.7 With an iron-sulfur protein as acceptor
 - 1.7.9 With a copper protein as acceptor
 - 1.7.99 With unknown physiological acceptors
 - 1.8 Acting on a sulfur group of donors
 - 1.9 Acting on a heme group of donors
 - 1.10 Acting on diphenols and related substances as donors
 - 1.11 Acting on a peroxide as acceptor

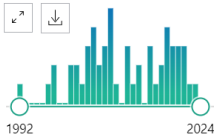
4)

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53 results

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RESULTS BY YEAR



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ARTICLE ATTRIBUTE

2 articles found by citation matching

TMβETADISC-RBF: Discrimination of beta-barrel membrane proteins using RBF networks and PSSM profiles.

Ou YY, et al. Comput Biol Chem. 2008. PMID: 18434251

Current developments on beta-barrel membrane proteins: sequence and structure analysis, discrimination and prediction.

Gromiha MM, et al. Curr Protein Pept Sci. 2007. PMID: 18220845 Review.

Show all

☐ OMPdb: A Global Hub of Beta-Barrel Outer Membrane Proteins.

1 Roumia AF, Tsigirgos KD, Theodoropoulou MC, Tamposis IA, Hamodrakas SJ, Bagos PG. Front Bioinform. 2021 Apr 9;1:646581. doi: 10.3389/fbinf.2021.646581. eCollection 2021.

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	PMID	Title	Authors	Citation	First Authc	Journal/Bc	Publication	Create Dat	PMCID	NIHMS ID	DOI			
2	8805593	Human dU Mol CD, H: Structure.	Mol CD	Structure	1996	#####					10.1016/s0969-2126(96)00114-1			
3	17557332	A consensi	Garrow AC	Proteins. 2	Garrow AC	Proteins	2007	#####			10.1002/prot.21439			
4	10210191	Structural	Juere DH, f	Protein Sci	Juere DH	Protein Sci	1999	#####	PMC2144101		10.1110/ps.8.1.122			
5	37432995	General fe	Montezani	Proc Natl /	Montezani	Proc Natl /	2023	#####	PMC10629564		10.1073/pnas.2220762120			
6	19380741	Single-nucl	Stoddart D	Proc Natl /	Stoddart D	Proc Natl /	2009	#####	PMC2683137		10.1073/pnas.0901054106			
7	1565623	Human au	Marchalor	Proc Natl /	Marchalor	Proc Natl /	1992	#####	PMC48859		10.1073/pnas.89.8.3325			
8	30153309	Select gp1	Lertjuthap	PLoS Path	Lertjuthap	PLoS Path	2018	#####	PMC6130882		10.1371/journal.ppat.1007278			
9	22073138	Protein do	Prakash A,	PLoS One.	Prakash A	PLoS One	2011	#####	PMC3206015		10.1371/journal.pone.0025570			
10	33137176	Novel ribo	Zatopek KI	Nucleic Ac	Zatopek KI	Nucleic Ac	2020	#####	PMC7708050		10.1093/nar/gkaa986			
11	20601684	Structure c	Nureki O,	Nucleic Ac	Nureki O	Nucleic Ac	2010	#####	PMC2978374		10.1093/nar/gkq605			
12														

8)h-index=106

citations= 54165



Burkhard Rost

Professor of Computer Science & Computational Biology & Bioinformatics, TUM, Munich

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machine learningartificial intelligencebioinformaticsmachine learningcomputer scienceprotein prediction

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YEAR

The transcriptional landscape of the mammalian genome

P Caminci, T Kasukawa, S Katayama, J Gough, MC Frith, N Maeda, ... science 309 (5740), 1559-1563

4017

2005

Prediction of protein secondary structure at better than 70% accuracy

B Rost, C Sander

Journal of molecular biology 232 (2), 584-599

3888

1993

Twilight zone of protein sequence alignments

B Rost

Protein engineering 12 (2), 85-94

2199

1999

Combining evolutionary information and neural networks to predict protein secondary structure

B Rost, C Sander

Proteins: Structure, Function, and Bioinformatics 19 (1), 55-72

1914

1994

The PredictProtein server

B Rost, G Yachdav, J Liu

Nucleic acids research 32 (suppl_2), W321-W326

1860

2004

PHD: Predicting one-dimensional protein structure by profile-based neural networks

B Rost

Methods in enzymology 266, 525-539

1720

1996

Cited by

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Citations

54185

13804

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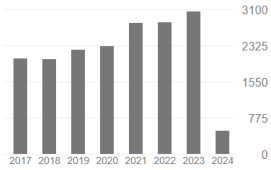
106

57

i10-index

271

182



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10 articles

128 articles

not available

available

Based on funding mandates

5)



Kihara D

Search

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219 results

Page 1 of 22

RESULTS BY YEAR



1997 2024

TEXT AVAILABILITY

- ☐ Abstract
☐ Free full text
☐ Full text

ARTICLE ATTRIBUTE

☐ Computational protein function predictions.

1 Kihara D.

Cite Methods. 2016 Jan 15;93:1-2. doi: 10.1016/j.jymeth.2016.01.001.
PMID: 26778120 No abstract available.

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☐ Computational prediction of disordered binding regions.

2 Basu S, Kihara D, Kurgan L.

Cite Comput Struct Biotechnol J. 2023 Feb 10;21:1487-1497. doi: 10.1016/j.csbj.2023.02.018. eCollection 2023.
PMID: 36851914 Free PMC article. Review.

Share

☐ Editorial: Expert opinions in protein bioinformatics: 2022.

3 Kihara D.

Cite Front Bioinform. 2024 Jan 5;3:1338560. doi: 10.3389/fbinf.2023.1338560. eCollection 2023.
PMID: 38350435 Free PMC article. No abstract available.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	PMID	Title	Authors	Citation	First Author	Journal/Book	Publication	Create Date	PMCID	NIHMS ID	DOI			
2	11802444	[Genome e	Kihara D, Tanpakush	Kihara D	Tanpakush	2001	#####							
3	9455244	[Internet r	Kihara D, K Tanpakush	Kihara D	Tanpakush	1997	#####							
4	28262392	Variability	Monroe L, Structure.	Monroe L	Structure	2017	#####	PM53821	NIHMS854	10.1016/j.str.2017.02.004				
5	32064000	2DKD: a to	DeVilLe JS, Source Co	DeVilLe JS	Source Co	2020	#####	PMC7011505		10.1186/s13029-020-0077-1				
6	31698087	Current pri	Karuppasa Semin Can	Karuppasa	Semin Can	2021	#####	PMC92335	NIHMS181	10.1016/j.semcan.2019.10.019				
7	25554786	Structure e	Liu Y, Shen Science. 2	Liu Y	Science	2015	#####	PMC43077	NIHMS656	10.1126/science.1261962				
8	18535227	The emerg	Hu JC, Arai Science. 2	Hu JC	Science	2008	#####			10.1126/science.320.5881.1289b				
9	33828153	Analyzing e	Jain A, Ter. Sci Rep. 2	Jain A	Sci Rep	2021	#####	PMC8027171		10.1038/s41598-021-87204-z				
10	31863054	A prospect	Chiba S, Ol Sci Rep. 2	Chiba S	Sci Rep	2019	#####	PMC6925144		10.1038/s41598-019-55069-y				
11	31784686	NNTox: Ge	Jain A, Kih Sci Rep. 2	Jain A	Sci Rep	2019	#####	PMC6884647		10.1038/s41598-019-54405-6				
12														

6)

Similar articles

Evolution. Tinkering inside the organelle.

Alcock F, Clements A, Webb C, Lithgow T.

Science. 2010 Feb 5;327(5966):649-50. doi: 10.1126/science.1182129.

PMID: 20133559 No abstract available.

Transport proteins (carriers) of mitochondria.

Wohlrab H.

IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.

PMID: 18816452 Review.

Systematic analysis of the twin cx(9)c protein family.

Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.

J Mol Biol. 2009 Oct 23;393(2):356-68. doi: 10.1016/j.jmb.2009.08.041. Epub 2009 Aug 21.

PMID: 19703468

Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.

Javadov S, Karmazyn M, Escobales N.

J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.

PMID: 19509316

A CaPful of mechanisms regulating the mitochondrial permeability transition.

Di Lisa F, Bernardi P.

J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.jmcc.2009.03.006. Epub 2009 Mar 19.

PMID: 19303419 Review.

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Leopardus geoffroyi
(Geoffroy's cat)

Search assembly
Location, gene or phenotype

Ideogram View

Unplaced/unlocalized scaffolds: 28

Assembly: O.geoffroyi.Qqe1_pat1.0 (GCF_1018350155.1) • Chr A1 (NC_059326.1) •

NC_059326.1: 1 - 239,694,388

NC_059326.1: 20 nt 40 nt 60 nt

Genes, NCBI Leopardus geoffroyi Annotation Release 100 - 100%
11EER PCH9 10FC 10FC

Genes, Ensembl Rapid release 2021-08
11EER PCH9 10FC 10FC

Chr A1 (NC_059326.1)
Chr C1 (NC_059328.1)
Chr D1 (NC_059329.1)
Chr E1 (NC_059330.1)
Chr A2 (NC_059331.1)
Chr B2 (NC_059332.1)
Chr C2 (NC_059333.1)
Chr D2 (NC_059334.1)
Chr E2 (NC_059335.1)
Chr A3 (NC_059336.1)
Chr B3 (NC_059337.1)
Chr C3 (NC_059338.1)
Chr D3 (NC_059339.1)
Chr E3 (NC_059340.1)
Chr B4 (NC_059341.1)
Chr D4 (NC_059342.1)
Chr X (NC_059343.1)
Chr MT (NC_028320.1)

Too many (2556) genes in the region to enable exon navigation.

Tools • Tracks • Download • 239,694,388

RNA-seq exon coverage, aggregate (filtered), NCBI Leopardus geoffroyi Annotation Release 100 - log base 2 scaled

RNA-seq intron-spanning reads, aggregate (filtered), NCBI Leopardus geoffroyi Annotation Release 100 - log base 2 scaled

Tracks and User Data

Eukaryota / Metazoa / Chordata / Mammalia / Rodentia / Caviidae / Cavia

Domestic guinea pig (*Cavia porcellus*) is a species of rodent in the family *Caviidae* (cavies).

[Browse taxonomy](#)

Current scientific name	<i>Cavia porcellus</i>
Common name	domestic guinea pig, guinea pig
Taxonomic rank	species
NCBI Taxonomy ID	10141

For more details see [NCBI Taxonomy](#)

Genome

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Image may not have been verified for accuracy by NCBI Taxonomy.

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Cavia porcellus
(domestic guinea pig)

Assembly: Cavpor3.0 (GCF_00015735.1)

Search assembly
Location, gene or phenotype

Unplaced/unlocalized scaffolds: 3,143

Ideogram View

Ideogram is not available
no assembled nuclear chromosomes

Non-nuclear chromosomes
MT

Tracks and User Data

BLAST

Tracks by Accession

History

NT_176419.1: 1 - 88,675,666

NT_176419.1 (unplaced)

Chromosomes
Chr MT (NC_000884.1)

Scaffolds (viewed)
NT_176419.1 (unplaced)

Scaffolds (largest)

NT_176390.1 (unplaced)
NT_176391.1 (unplaced)
NT_176392.1 (unplaced)
NT_176393.1 (unplaced)
NT_176394.1 (unplaced)
NT_176395.1 (unplaced)
NT_176396.1 (unplaced)
NT_176397.1 (unplaced)
NT_176398.1 (unplaced)
NT_176399.1 (unplaced)
NT_176400.1 (unplaced)
NT_176401.1 (unplaced)
NT_176402.1 (unplaced)
NT_176403.1 (unplaced)
NT_176404.1 (unplaced)
NT_176405.1 (unplaced)
NT_176406.1 (unplaced)

Genes, NCBI Cavia porcellus Annotation Release 104
Genes, Ensembl release 111

RNA-seq exon coverage, aggregate (filtered)
RNA-seq intron-spanning reads, aggregate (filtered), NCBI Cavia porcellus Annotation Release 103 - log base 2 scaled

for: There are too many (684) genes in the region.
arrow the region to enable exon navigation.

Eukaryota / Metazoa / Chordata / Mammalia / Carnivora / Canidae / Canis / Canis lupus

Canis lupus familiaris ☆

Dog (*Canis lupus familiaris*) is a subspecies of gray wolf (*Canis lupus*).

[Browse taxonomy](#)

Current scientific name	<i>Canis lupus familiaris</i>
Common name	dog, dogs
Taxonomic rank	subspecies
NCBI Taxonomy ID	9615

For more details see [NCBI Taxonomy](#)



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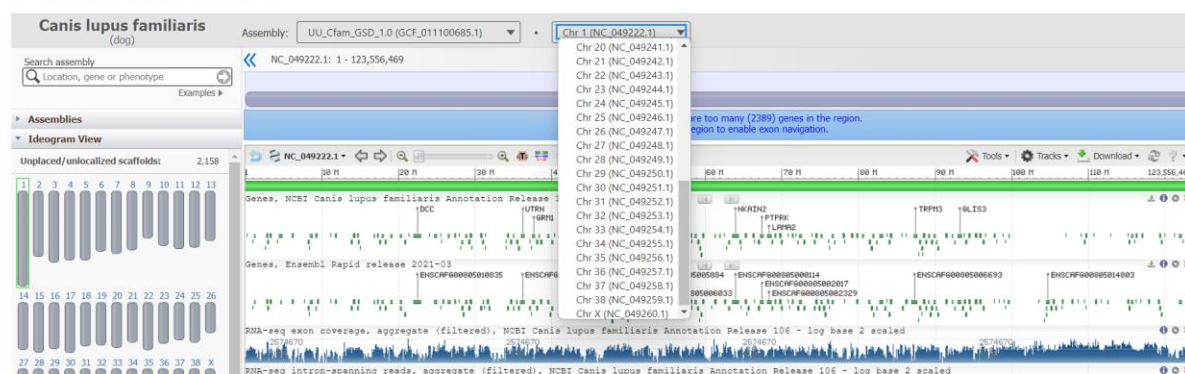
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Genome Data Viewer



Eukaryota / Viridiplantae / Streptophyta / Magnoliopsida / Brassicales / Brassicaceae / Arabidopsis

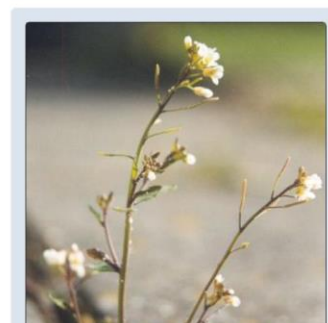
Arabidopsis thaliana ☆

Thale cress (*Arabidopsis thaliana*) is a species of eudicot in the family *Brassicaceae* (mustard family) that is widely used as an experimental model organism.

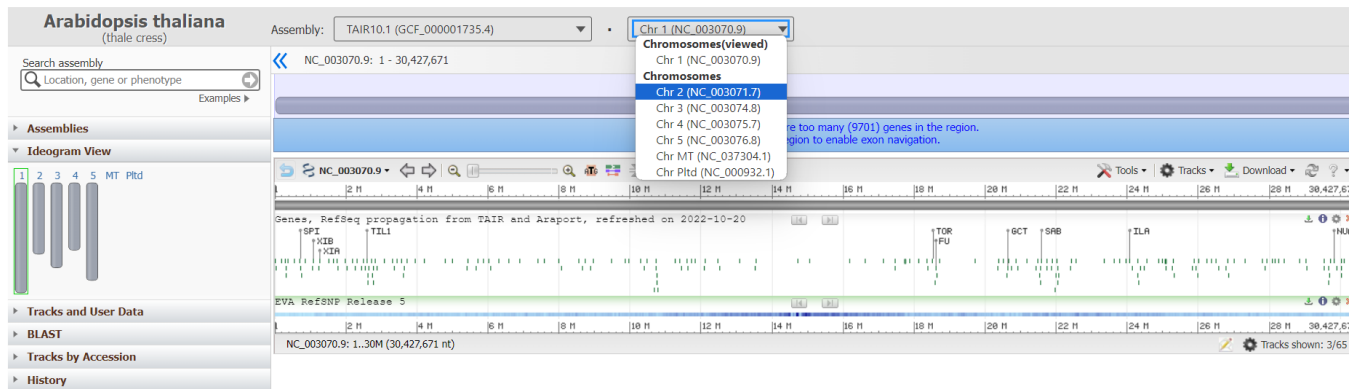
[Browse taxonomy](#)

Current scientific name	<i>Arabidopsis thaliana</i>
Common name	thale cress, mouse-ear cress, thale-cress
Taxonomic rank	species
NCBI Taxonomy ID	3702

For more details see [NCBI Taxonomy](#)



Genome Data Viewer



Compiling the above images:

1. **Human:**
 - Scientific Name: Homo sapiens
 - Taxonomy ID: 9606
 - Number of Chromosomes: 23 pairs (46 chromosomes)
2. **Cat:**
 - Scientific Name: Felis catus
 - Taxonomy ID: 9685
 - Number of Chromosomes: 19 pairs (38 chromosomes)
3. **Dog:**
 - Scientific Name: Canis lupus familiaris
 - Taxonomy ID: 9615
 - Number of Chromosomes: 19 pairs (38 chromosomes)
4. **Domestic Guinea Pig:**
 - Scientific Name: Cavia porcellus
 - Taxonomy ID: 10141
 - Number of Chromosomes: 16 pairs (32 chromosomes)
5. **Thale Cress (Arabidopsis thaliana):**
 - Scientific Name: Arabidopsis thaliana
 - Taxonomy ID: 3702
 - Number of Chromosomes: 5 pairs (10 chromosomes)

racemase, 5.1.1.3, P56868 [advanced](#)

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Asparagine synthase (glutamine-hydrolysing)

Asparagine synthetase B catalyses the ATP dependent formation of asparagine from aspartate using either glutamine or ammonia as a nitrogen source. The enzyme is a homodimer with two active sites [PMID:10587437] in each subunit, one responsible for glutamine hydrolysis and one for asparagine synthesis, separated by a hydrophobic intramolecular tunnel around 20Å long [PMID:12706338]. Sequencing analysis has shown the synthetase to belong to the larger glutamine transferase family.

Reference Protein and Structure

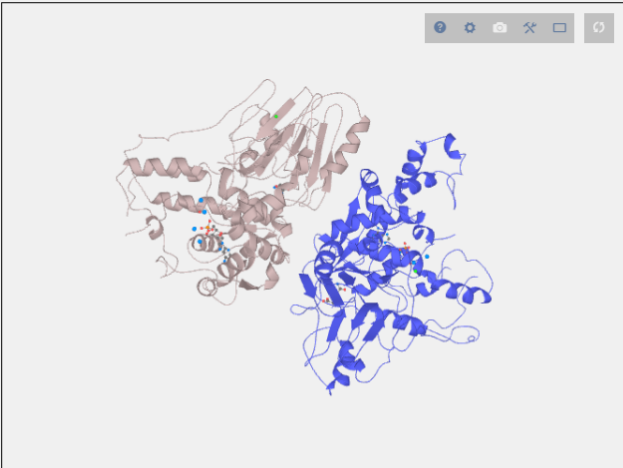
Sequence
[P22106](#) [UniProt](#) (6.3.5.4) [InterPro](#) (Sequence Homologues) (PDB Homologues)

Biological species
[Escherichia coli K-12](#) (Bacteria) [UniProt](#)

PDB
[1ct9](#) - CRYSTAL STRUCTURE OF ASPARAGINE SYNTHETASE B FROM ESCHERICHIA COLI (2.0 Å) [PDB](#)

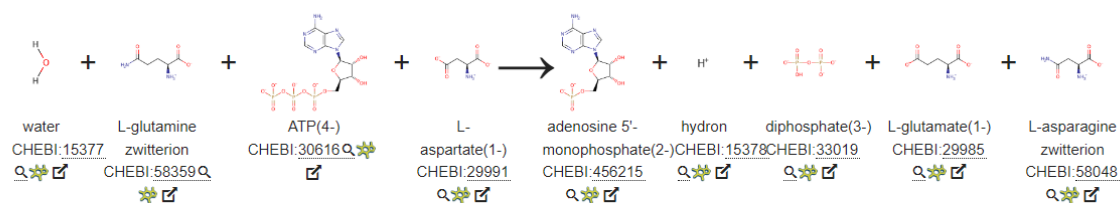
Catalytic CATH Domains
[3.60.20.10](#) [3.40.50.620](#) (see all for [1ct9](#))

Cofactors
 Magnesium(2+) (1) [Metal MACiE](#)



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Enzyme Reaction (EC:6.3.5.4Q)



Enzyme reaction links: [IntEnz](#) [ENZYME](#) [ExplorEnz](#) [Rhea](#) [BRENDA](#) [KEGG](#) [METACyc](#)

Alternative enzyme names: Asparagine synthetase (glutamine-hydrolyzing), Glutamine-dependent asparagine synthetase, Asparagine synthetase B, AS, AS-B,

Enzyme Mechanism

Mechanism Proposal 1 ☆☆☆

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Introduction

The enzyme catalyses three distinct chemical reactions: glutamine hydrolysis to yield ammonia (which is then channelled to the second active site) takes place in the N-terminal domain. The C-terminal active site mediates both the synthesis of a beta-aspartyl-AMP intermediate and its subsequent reaction with ammonia. However, the exact order of the partial reactions is still somewhat unclear [PMID:9748330, PMID:12706338], here we show the hydrolysis of the glutamate occurring first.

Glutamine hydrolysis occurs at the N terminal. A nucleophilic Cys residue attacks the glutamine substrate, displacing ammonia, forming a substrate-

Catalytic Residues Roles

UniProt	PDB* (1ct9)		
Cys2 (N-term)	Ala1A (N-term)	Acts as a general acid/base to activate the cysteine nucleophile.	proton acceptor, proton donor
Leu51 (main-C)	Leu50A (main-C)	Helps stabilise the reactive intermediates formed.	hydrogen bond acceptor, electrostatic stabiliser
Thr322, Arg325	Thr321A, Arg324A	Bind and stabilise the phosphate groups of the ATP and reactive intermediates formed.	hydrogen bond donor, electrostatic stabiliser
Cys2	Ala1A	Acts as a catalytic nucleophile in the glutaminase domain reaction.	covalently attached, hydrogen bond acceptor, nucleofuge, nucleophile, proton acceptor, proton donor
Gly76 (main-N), Asn75	Gly75A (main-N), Asn74A	Forms the oxyanion hole that stabilises the reactive intermediates and transition states formed.	hydrogen bond donor, electrostatic stabiliser

*PDB label guide - RESx(y)B(C) - RES: Residue Name; x: Residue ID in PDB file; y: Residue ID in PDB sequence if different from PDB file; B: PDB Chain; C: Biological Assembly Chain if different from PDB. If label is "Not Found" it means this residue is not found in the reference PDB.

Chemical Components

proton transfer, bimolecular nucleophilic addition, proton relay, enzyme-substrate complex formation, overall reactant used, intermediate formation, unimolecular elimination by the conjugate base, enzyme-substrate complex cleavage, deamination, intermediate collapse, overall product formed, native state of enzyme regenerated

References

1. Tesson AR *et al.* (2003), *Arch Biochem Biophys*, **413**, 23-31. Revisiting the steady state kinetic mechanism of glutamine-dependent asparagine synthetase from *Escherichia coli*. DOI:10.1016/s0003-9861(03)00118-8. PMID:12706338.
2. Fresquet V *et al.* (2004), *Bioorg Chem*, **32**, 63-75. Kinetic mechanism of asparagine synthetase from *Vibrio cholerae*. DOI:10.1016/j.bioorg.2003.10.002. PMID:14990305.
3. Larsen TM *et al.* (2000), *Biochemistry*, **39**, 7330-. Three-dimensional structure of escherichia coli asparagine synthetase B: A short journey from substrate to product. PMID:10852734.
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 3 [No authors listed]
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	A	B	C	D	E	F	G	H	I	J	K	L	M
1	PMID	Title	Authors	Citation	First Author	Journal/Book	Publication	Create Date	PMCID	NIHMS ID	DOI		
2	38396303	Author Co	Trinh TH, V	Nature. 20	Trinh TH	Nature	2024	#####			10.1038/s41586-024-07115-7		
3	38396101	Machine learning rev	Nature. 2024	Feb 23.	Nature	2024	#####				10.1038/d41586-024-00431-y		
4	38396100	The first bulk ceramic	Nature. 2024	Feb 23.	Nature	2024	#####				10.1038/d41586-024-00443-8		
5	38396099	'All of Us' i	Kozlov M.	Nature. 20	Kozlov M	Nature	2024	#####			10.1038/d41586-024-00568-w		
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10	38396094	CAR-T ther	Mullard A.	Nature. 20	Mullard A	Nature	2024	#####			10.1038/d41586-024-00470-5		
11	38396093	First privat	Witze A.	Nature. 20	Witze A	Nature	2024	#####			10.1038/d41586-024-00549-z		

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```

In [2]: import requests

def fetch_fasta_record(db, accession):
    base_url = "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi"
    params = {
        'db': db,
        'rettype': 'fasta',
        'id': accession,
    }

    response = requests.get(base_url, params=params)

    if response.status_code == 200:
        return response.text
    else:
        return f"Error: {response.status_code} - {response.text}"

# Example usage:
database_name = "nucleotide"
accession_number = "NM_00520"
fasta_sequence = fetch_fasta_record(database_name, accession_number)

print(fasta_sequence)

>NM_00520.6 Homo sapiens hexosaminidase subunit alpha (HEXA), transcript variant 2, mRNA
CTCAGTGGCAGCCCCCTCGAGAGGGGAGACAGCGGGCCATGACAAGCTCCAGGCTTTGGTTTTTCGC
TGCTGCTGGCGGACGGTTCGAGGACGGGCGACGGCCCTTGCCCTCGGCTCAGAACTTCCAAACCTC
CGACCAAGCGCTACGCTCTTTACCGGAACAATCTCAATTCAGTAGCATGTCAAGCTCGGCGCGGACGCC
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GGACAGGATCTCCGTGATTAAGTCTGCTACTCTGCTGCTGAGGCTTGGCAGCTTTGGACAG
TGAATCCAGTCTCAATAATACCTATGAGTTCATGAGCACATTCTTCTTGAAGAGTCACTGTCTTCC
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GTGCTTGGAGAGAAAGGGGCGGCTGCTGCGCTCGCATCAATAAAGAGTAAATGAGGATTTTCTATAA
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TTTGTGAGGAAATGAGTGTGTGTGAGGCTGGGCTGCTGAGAGGGTCTTGTGAGTGTGATTT

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BindingDB has accelerated collection of COVID related data. You can find the result [here](#).

Try the BindingDB Browser Extension

We are excited to share our new browser extension, **BDBFind**. Once installed in your browser, BDBFind automatically lets you know when BindingDB has the data from an article, PubMed Entry, or US Patent you are looking at online and provides direct links to view or download the data. Get BDBFind by searching for it in the Chrome webstore or in Firefox extensions or by following these links:

<https://chrome.google.com/webstore/search/bdbfind> (Once installed, click on the "jigsaw puzzle" icon to keep the BDB icon displayed.)

<https://addons.mozilla.org/addon/bdbfind>.

Watch BDBFind in action in this 1.5 minute video:

Cancer genes

General polymorphism databases

Cancer gene databases

ArrayMap
BCCTBbp
BreCAN-DB
Cancer RNA-Seq Nexus
Cancer3D
CancerGenes
CancerPPD
Candidate Cancer Gene Database
CanGEM
CanSAR
CaSNP
cBioPortal
CCDB
ccmGDB
CellLineNavigator
ChimerDB
CIVICdb
ClinVar
CMPD
Colorectal Cancer Atlas
COLT-Cancer
COSMIC
CTDatabase
dbDEMC
dbDEPC
DIVAS
DriverDBv2
FusionCancer
HLungDB
HPTAA
IARC TP53 Database
IGDB.NSCLC
intoGen
ITTACA
Inc2cancer
MethHC
MethyCancer
MoKCa
MutationAligner
Network of Cancer Genes
